

Bagged Random Subset Methods for Forecast Combination: Theory, Monte Carlo, and Empirical Evidence

Exact Results, Power-Law Excess Risk, and the Cube-Root Law

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Abstract

We study Random Subset Methods (RSM) for forecast combination in a panel of m individual forecasters. The paper proceeds inductively. We first establish exact population results under the common-loading covariance model: a *conservation law* governing how precision-weighted RSM weights redistribute across subsets, an exact formula for the infinite-bagged excess risk $R_k^{\text{bag},\infty}$, monotone decrease of $R_k^{\text{bag},\infty}$ in k (proved under a stated contraction condition, verified numerically), and a first-order condition for the optimal subset size k^* . We then show that the average-subset risk is tractable via Taylor expansion, yielding a square-root optimal subset size $k_{\text{sub}}^* \propto T^{1/2}$. The infinite-bagged risk resists the same treatment because the key conditional expectation has no closed form without further restrictions. We use Monte Carlo experiments with known Σ_e to characterise the functional form of $R_k^{\text{bag},\infty}$ empirically: the power-law family $R_k^{\text{bag},\infty} \approx C z_k^\alpha$, $z_k = 1/k - 1/m$, fits with $R^2 > 0.97$ in all designs. The exponent $\hat{\alpha} \in (1, 2)$ is determined by the heterogeneity ratio H of idiosyncratic variances and implies an optimal subset size $k^* \propto T^{1/(\alpha+1)}$ scaling between $T^{1/3}$ (small heterogeneity, $\alpha \rightarrow 2$) and $T^{1/2}$ (dominant forecaster, $\alpha \rightarrow 1$). For the endogenous case (estimated Σ_e) we provide a tractable result under strict Gaussianity and common loading via the Kan–Zhou correction, which preserves the $T^{1/(\alpha+1)}$ scaling exponent; the subset-level correction factor $(T-2)/(T-k-2)$ is small for $k \ll T$ and produces only a modest level shift in k^* . Empirically the endogenous optimal k^* is substantially larger than the tractable prediction, driven by non-common-loading misspecification rather than estimation noise per se. Applying RSM to FRED-MD GDP forecasts yields statistically significant RMSFE reductions of 10–30% over the equal-weighted mean. The Survey of Professional Forecasters—a near-homogeneous panel—confirms that RSM offers little advantage when forecasters are nearly identical in quality.

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1 Scope and Main Message

The central question is: in a large panel of m individual forecasters, how many should a practitioner randomly subsample, and how should subsample forecasts be combined, to minimise out-of-sample loss?

Two distinct RSM objects have been studied in the literature. The *average-subset estimator* picks a random size- k subset, fits constrained OLS within it, and reports that single forecast. The *bagged estimator* averages G such draws; as $G \rightarrow \infty$ it converges to the infinite-bagged estimator. Since $w \mapsto w' \Sigma_e w$ is convex, Jensen's inequality gives $Q_k^{\text{bag}, \infty} \leq Q_k^{\text{sub}}$, so bagging always weakly reduces population risk. Yet prior work has largely analysed Q_k^{sub} and assumed a functional form for the bagged excess risk. We derive what can be derived exactly, then use Monte Carlo to characterise what cannot.

Logic of the paper. Sections 2–3 establish the framework and show that the equal-weighted mean and full estimated optimal are special cases of RSM. Section 4 defines RSM and derives the exact finite-bagging bridge. Section 5 exploits the common-loading structure to obtain the conservation law, exact excess-risk formula, monotone decrease of $R_k^{\text{bag}, \infty}$, and a first-order condition for k^* . Section 6 derives the average-subset optimum and shows why the bagged optimum resists a parallel derivation. Section 7 uses Monte Carlo with known Σ_e to motivate the power-law parametrisation of $R_k^{\text{bag}, \infty}$, characterise $\hat{\alpha}$, and validate the implied k^* . Section 8 studies the endogenous case, beginning with a tractable result under strict Gaussianity and closing with Monte Carlo evidence for general designs. Sections 9–10 present the empirical applications.

1.1 Motivation and Relation to the Literature

The forecast combination puzzle and Elliott–Liao (2025). A long-standing puzzle in forecast combination is that the equal-weighted mean of individual forecasts typically outperforms the estimated minimum-variance combination, despite the latter being the solution to a simple constrained OLS problem [??]. ? provide a complementary perspective: rather than attributing the puzzle solely to estimation error, they ask how large the *population* gains from optimal combination can be under empirically motivated restrictions on the forecast-error covariance $\tilde{\Sigma}$. Their model,

$$\tilde{\Omega} = \sigma_\varepsilon^2 \boldsymbol{\iota} \boldsymbol{\iota}' + \tilde{\Sigma}, \quad (1)$$

is precisely our common-loading structure ($q = \sigma_\varepsilon^2$, $D = \tilde{\Sigma}$). Using SPF data to discipline the space of $\tilde{\Sigma}$, they show that for most empirically plausible parameterisations the population gain $Q^{\text{ave}} - Q^{\text{opt}}$ is small relative to estimation error $(m - 1)/T$ — providing a theoretical rationale for the combination puzzle that operates even in population, before

any finite-sample noise.

Inflation factors: Elliott–Liao vs RSM. A key ingredient in the puzzle is the *inflation factor* — the multiplier by which the in-sample optimal-combination loss exceeds its true population value because the weights are estimated rather than known. For the full m -weight optimal combination estimated by constrained OLS, the standard degrees-of-freedom correction is

$$I_m(T) = \frac{T-1}{T-m}, \quad (2)$$

the exact Gaussian correction for estimating $m-1$ free parameters from T observations. Its leading-order excess over 1 is

$$I_m(T) - 1 \approx \frac{m-1}{T}, \quad (3)$$

and it is this quantity $(m-1)/T$ that ? use as their benchmark for the expected estimation cost. They call it the *best case* for OLS because it is the minimum possible estimation error: it assumes Gaussian errors, a correctly specified model, and the full m -dimensional weight vector. Any real-world estimation procedure can only do worse. Their criterion for full optimal combination to be worthwhile is then $Q^{\text{ave}} - Q^{\text{opt}} > (m-1)/T \cdot (\sigma_\varepsilon^2 + Q^{\text{opt}})$; for most empirically calibrated $\tilde{\Sigma}$ this condition fails, which they show as the rationale for the combination puzzle.

The present paper departs from this framework in a single but critical respect: we use the inflation factor $(T-1)/(T-k)$ for a *subset* of size $k < m$, not the full-model factor $(T-1)/(T-m)$. Since $k \ll m$, the RSM inflation is substantially smaller:

$$\frac{T-1}{T-k} \ll \frac{T-1}{T-m}, \quad \text{i.e.} \quad \frac{k-1}{T} \ll \frac{m-1}{T}. \quad (4)$$

This opens a window of viable gains that the full-model approach cannot access: RSM can produce a net positive out-of-sample improvement even when Elliott–Liao’s benchmark says full OLS combination is not worth estimating, because RSM’s estimation cost is only $(k-1)/T$ rather than $(m-1)/T$. Choosing k optimally therefore means choosing the largest subset for which the RSM gain $Q^{\text{ave}} - Q_k^{\text{bag},\infty}$ exceeds the RSM-specific inflation cost — the central problem this paper solves.

Their Proposition 1 gives a precise characterisation: when $\boldsymbol{\iota}$ is an eigenvector of $\tilde{\Sigma}$ (equivalently, all row sums of $\tilde{\Sigma}$ are equal), the equal-weighted mean *is* the optimal combination and gains are exactly zero. In our notation this corresponds to $V_\delta = 0$, hence $C_\delta = 0$, the near-homogeneous case in which $k^* = 1$ (Remark 6).

Their key positive result (Section 3.2, Proposition 2) concerns the case where gains *are* large. They characterise the worst-case $\tilde{\Sigma}$ for the equal-weighted mean: the maximiser of $Q^{\text{ave}} - Q^{\text{opt}}$ over the restricted space has a block structure $\Sigma_{11} = I_{m_1}$, $\Sigma_{12} = m_1^{-1} \boldsymbol{\iota}_1 \boldsymbol{\iota}_2'$, and

the corresponding optimal weights are

$$w^{\text{opt}} = \begin{pmatrix} (1/m_1) \iota_1 \\ 0 \end{pmatrix}. \quad (5)$$

That is: *take the equal-weighted average of the first m_1 forecasters and discard the remaining $m - m_1$* . This is the Complete Subset estimator of ? (CSR): a fixed subset of size m_1 with equal within-subset weights. Elliott and Liao thus establish that CSR is the *optimal* response when gains from full optimal combination are largest — but they stop short of two important questions: (i) which forecasters should be in the retained subset, and (ii) how large should the subset be?

How RSM improves on the Elliott–Liao prescription. RSM addresses both gaps and goes one step further.

- (i) **Subset selection via randomisation.** Searching deterministically for the best size- m_1 subset is combinatorially intractable ($\binom{m}{m_1}$ candidates). RSM replaces this with uniform random draws and averaging — eliminating subset-selection variance without any search.
- (ii) **Optimal within-subset weights.** Elliott–Liao’s prescription uses equal weights within the retained subset. RSM instead applies constrained minimum-variance OLS within each draw, giving weights $w_{S,i} = \tau_i/A_S$. Since $w_S = \arg \min_{w' \iota = 1} w' \Sigma_S w$, we have $w'_S \Sigma_S w_S \leq (1/k^2) \iota' \Sigma_S \iota$ — optimal weights are always at least as good as equal weights within the same subset. Taking expectations over S : $Q_k^{\text{RSM}} \leq Q_k^{\text{CSR}}$ strictly whenever Σ_e is not a multiple of the identity.
- (iii) **Optimal k^* .** Neither Elliott–Liao nor CSR provide a theory for the optimal subset size. The present paper derives this: $k^* \propto T^{1/(\alpha+1)}$ under the power-law parametrisation of the excess risk, with the endogenous approximation $k^* \approx \lfloor \sqrt{T} \rfloor$ for practical use.

Chen and Maung (2023) and the cost of ad hoc k selection. ? develop nonparametric time-varying forecast combination for high-dimensional data. In their empirical application to inflation and unemployment forecasting they include CSR [?] as a benchmark, evaluated at $k \in \{1, 2, 3, 5\}$. Their Table 3 reveals a telling pattern (ASCFE ratios relative to the AR benchmark; below 1 is better):

Series	$k = 1$	$k = 2$	$k = 3$	$k = 5$
Inflation	0.98	0.95	0.93	0.96
Unemployment	1.01	1.02	1.01	0.97

For *unemployment*, performance is monotonically improving through $k = 5$ — the curve is still decreasing at the largest k they consider. For *inflation*, the minimum is near

$k = 3\text{--}4$ but was never precisely located; Chen and Maung do not study $k > 5$. In both cases they were on the decreasing side of the U-shaped $\text{RMSFE}(k)$ curve and never found the optimum. The theory in this paper predicts $k^* \approx \lfloor \sqrt{T} \rfloor \approx 10\text{--}12$ at the sample sizes they used ($T \approx 100\text{--}150$), substantially above $k = 5$.

Relation to Bayesian Model Averaging (BMA). Bayesian Model Averaging [??] forms the combined forecast as $\hat{y}^{\text{BMA}} = \sum_j P(M_j|y) \hat{y}_{M_j}$, where models are weighted by their posterior probabilities. In the forecast-combination context, “models” are typically individual forecasters or subsets of forecasters, and posteriors are proportional to the marginal likelihood (often approximated by BIC).

RSM and BMA share the structure of averaging over subsets, but differ in three fundamental ways.

(a) Prior and weighting. BMA assigns posterior-probability weights to subsets, concentrating mass on subsets with high predictive accuracy. RSM uses a *uniform* (flat) prior over all size- k subsets, so every subset receives the same drawing probability $1/\binom{m}{k}$. Under a flat prior over subsets of size k and equal within-subset weights, the BMA average equals the CSR estimator of ?. RSM replaces equal with optimal within-subset weights, strictly improving on this version of BMA.

(b) Within-subset treatment. BMA typically evaluates each model/subset by its marginal likelihood, treating within-subset weights as either equal or MLE-estimated. RSM explicitly minimises within-subset population variance, giving weights $w_{S,i} = \tau_i/A_S$ that are optimal in the mean-squared-error sense. The infinite-bagged RSM weight $\bar{v}_{k,i} = \mathbb{E}_S[\tau_i/A_S \cdot \mathbf{1}_{i \in S}]$ is therefore a *precision-weighted* average of subset memberships, not a likelihood-weighted average of model performances.

(c) Asymptotic behaviour. BMA posteriors concentrate on the “true” model as $T \rightarrow \infty$ (Bernstein–von Mises), becoming degenerate and eventually selecting a single forecaster. RSM with $k \rightarrow m$ converges to the full minimum-variance combination w^{opt} , never collapsing to a single model. RSM’s behaviour as k grows is governed by the excess risk $R_k^{\text{bag},\infty} \rightarrow 0$, which is fundamentally a frequentist variance-reduction argument rather than a Bayesian consistency result. The optimal k^* derived in this paper is therefore a frequentist bias-variance trade-off, with no natural Bayesian counterpart.

Position of this paper. We formalise RSM as the solution to a well-posed minimum-variance problem, derive exact population results, use Monte Carlo to characterise the functional form of the excess risk, and provide an actionable rule for k^* selection. Relative to Elliott–Liao, we show that RSM with optimal within-subset weights strictly dominates their CSR prescription and derive the missing k^* theory. Relative to Chen–Maung, we provide the theory that explains their empirical U-shape and guides the choice of k . Relative to BMA, RSM is a frequentist minimum-variance estimator with a natural flat-

prior interpretation and a tractable optimality theory.

2 Forecast-Combination Setup

Assumption 1 (Forecast model). *There are m individual forecasts $f_t = (f_{1t}, \dots, f_{mt})'$ of a scalar target y_t . Forecast errors $u_{it} = f_{it} - y_t$ satisfy*

$$\mathbb{E}(u_t) = 0, \quad \text{Var}(u_t) = \Sigma_e \succ 0, \quad \mathbb{E}(u_{it}u_{jt'}) = 0 \text{ for } t \neq t'.$$

For any weight vector w with $w'\boldsymbol{\iota} = 1$ where $\boldsymbol{\iota} = (1, \dots, 1)'$, the forecast $w'f_t$ has mean-squared prediction error

$$\text{MSPE}(w) = w'\Sigma_e w; :=; Q(w). \quad (6)$$

The scalar $s = \text{Var}(y_t - \mathbb{E}[y_t|f_t])$ is the irreducible forecast uncertainty shared by all combinations and plays no role in choosing w . We therefore focus on $Q(w)$ throughout.

3 Benchmarks, and Why Both Are Special Cases of RSM

3.1 Equal-Weighted Mean

The equal-weighted combination assigns weight $1/m$ to every forecaster:

$$w^{\text{ave}} = \frac{1}{m}\boldsymbol{\iota}, \quad Q^{\text{ave}} = \frac{1}{m^2}\boldsymbol{\iota}'\Sigma_e\boldsymbol{\iota}. \quad (7)$$

Since no parameters are estimated, Q^{ave} is the population MSPE with no finite-sample inflation.

3.2 Full Optimal Weights

The infeasible minimum-variance combination is

$$w^{\text{opt}} = \frac{\Sigma_e^{-1}\boldsymbol{\iota}}{\boldsymbol{\iota}'\Sigma_e^{-1}\boldsymbol{\iota}}, \quad Q^{\text{opt}} = (\boldsymbol{\iota}'\Sigma_e^{-1}\boldsymbol{\iota})^{-1}. \quad (8)$$

When Σ_e is estimated from T observations, the degrees-of-freedom inflation of the constrained OLS estimator gives

$$L^{\text{full}} \approx (s + Q^{\text{opt}})\frac{T-1}{T-m}, \quad (9)$$

which diverges as $m \rightarrow T$, reflecting the curse of dimensionality.

3.3 RSM Unifies Both Benchmarks

We defer the formal definition to Section 4, but note immediately the two limiting cases that motivate RSM as the natural unifying object:

- **Full pool** ($k = m$): the only subset of size m is the full panel, so the RSM weight equals w^{opt} and $Q_m^{\text{bag},\infty} = Q^{\text{opt}}$.
- **Homogeneous forecasters** ($\Sigma_e = \sigma^2[q\boldsymbol{\mu}' + I_m]$ up to a scale): within any subset the constrained OLS weights are equal, so the bagged weight converges to w^{ave} and $Q_k^{\text{bag},\infty} = Q^{\text{ave}}$ for all k .

For $1 \leq k < m$ and non-trivial heterogeneity, RSM interpolates between the two, strictly improving on Q^{ave} while avoiding the curse of dimensionality that makes L^{full} large when m is close to T .

4 RSM: Definition and the Exact Bagging Bridge

Definition 1 (RSM). Let $S \subset \{1, \dots, m\}$ be a random subset of size k , drawn uniformly without replacement from $\binom{m}{k}$ possible subsets. Let $\Sigma_{e,S}$ denote the $k \times k$ principal submatrix of Σ_e indexed by S . The *RSM weight vector for subset S* , embedded in $\mathbb{R}^m$, is

$$v_S = P_S \frac{\Sigma_{e,S}^{-1} \boldsymbol{\mu}}{\boldsymbol{\mu}' \Sigma_{e,S}^{-1} \boldsymbol{\mu}}, \quad (10)$$

where P_S places the k computed weights in their original coordinates and sets the remaining $m - k$ coordinates to zero. The *average-subset* forecast uses a single draw S . The *finite-bagged* forecast with G draws uses $\hat{v}_{k,G} = G^{-1} \sum_{g=1}^G v_{S_g}$, and the *infinite-bagged* weight is

$$\bar{v}_k = \mathbb{E}_S[v_S], \quad (11)$$

where the expectation is over the uniform distribution on subsets of size k .

The three population risks are

$$Q_k^{\text{sub}} = \mathbb{E}_S[v_S' \Sigma_e v_S], \quad (12)$$

$$Q_k^{\text{bag},G} = \mathbb{E}[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}], \quad (13)$$

$$Q_k^{\text{bag},\infty} = \bar{v}_k' \Sigma_e \bar{v}_k. \quad (14)$$

Proposition 1 (Exact finite-bagging bridge). *For G independent draws $S_1, \dots, S_G$,*

$$Q_k^{\text{bag},G} = Q_k^{\text{bag},\infty} + \frac{1}{G} (Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}), \quad (15)$$

and hence $Q_k^{\text{bag},\infty} \leq Q_k^{\text{bag},G} \leq Q_k^{\text{sub}}$ for all $G \geq 1$.

Proof. Let $V = v_S$ and $\bar{v} = \mathbb{E}[V]$. For G i.i.d. draws, $\hat{v}_{k,G} = G^{-1} \sum_g V_g$. Because $\Sigma_e \succ 0$,

$$\begin{aligned} \mathbb{E}[\hat{v}_{k,G}' \Sigma_e \hat{v}_{k,G}] &= \bar{v}' \Sigma_e \bar{v} + \mathbb{E}[(\hat{v}_{k,G} - \bar{v})' \Sigma_e (\hat{v}_{k,G} - \bar{v})] \\ &= Q_k^{\text{bag},\infty} + \frac{1}{G} \mathbb{E}[(V - \bar{v})' \Sigma_e (V - \bar{v})]. \end{aligned}$$

Expanding $\mathbb{E}[(V - \bar{v})' \Sigma_e (V - \bar{v})] = \mathbb{E}[V' \Sigma_e V] - \bar{v}' \Sigma_e \bar{v} = Q_k^{\text{sub}} - Q_k^{\text{bag},\infty}$ gives (15). $\square$

Remark 1. Identity (15) is exact: no distributional assumption on $\{f_t\}$ and no approximation. It shows that increasing G monotonically reduces population risk until the infinite-bagged limit is reached, and that the benefit of bagging equals $G^{-1}(Q_k^{\text{sub}} - Q_k^{\text{bag},\infty})$.

5 Common-Loading Structure: Exact Theory

5.1 Model and Notation

Assumption 2 (Common-loading covariance). *The forecast-error covariance matrix satisfies*

$$\Sigma_e = q \boldsymbol{\mathcal{U}}' + D, \quad D = \text{diag}(\sigma_1^2, \dots, \sigma_m^2), \quad q \geq 0, \sigma_i^2 > 0. \quad (16)$$

Define precision weights $\tau_i = \sigma_i^{-2}$, $A_S = \sum_{i \in S} \tau_i$, and $A_m = \sum_{i=1}^m \tau_i$.

Under this model, for any subset S of size k , the Woodbury identity yields $\Sigma_{e,S}^{-1} = D_S^{-1} - q(1 + qA_S)^{-1} \boldsymbol{\tau}_S \boldsymbol{\tau}_S'$, where $\boldsymbol{\tau}_S = (\tau_i)_{i \in S}$. The constrained optimal weight within S is

$$w_{S,i} = \frac{\tau_i}{A_S}, \quad i \in S, \quad (17)$$

which does not depend on the common-loading parameter q . Consequently the subset risk is

$$Q(w_S) = q + \frac{1}{A_S}; \quad Q(S). \quad (18)$$

The oracle risk and equal-weight risk are

$$Q^{\text{opt}} = q + \frac{1}{A_m}, \quad Q^{\text{ave}} = q + \frac{\sum_i \tau_i^{-1}}{m^2}. \quad (19)$$

These expressions can be written more compactly in terms of the two scalar parameters used throughout the Monte Carlo experiments:

$$B := \frac{1}{\bar{\tau}} = \frac{m}{A_m}, \quad H := \overline{\sigma^2} \bar{\tau}, \quad (20)$$

where $\bar{\tau} = A_m/m$ is the mean precision and $\overline{\sigma^2} = m^{-1} \sum_i \sigma_i^2$ is the mean idiosyncratic variance. B is the mean idiosyncratic variance (the common scale of the σ_i^2) and $H \geq 1$

measures how spread out the σ_i^2 are ($H = 1$ iff all σ_i^2 are equal). Substituting $\sum_i \tau_i^{-1} = m \bar{\sigma}^2 = mHB$ gives

$$Q^{\text{opt}} = q + \frac{B}{m}, \quad Q^{\text{ave}} = q + \frac{HB}{m}, \quad Q^{\text{ave}} - Q^{\text{opt}} = \frac{B(H-1)}{m}. \quad (21)$$

The inequality $H \geq 1$ holds because the AM–HM inequality applied to $\{\sigma_i^2\}$ gives $\bar{\sigma}^2 \geq m / \sum_i \tau_i = 1/\bar{\tau}$, so $H = \bar{\sigma}^2 \bar{\tau} \geq 1$, with equality iff all σ_i^2 are equal. Hence $Q^{\text{ave}} \geq Q^{\text{opt}}$ always, with equality only in the homogeneous case.

5.2 Infinite-Bagged Weights and the Conservation Law

For $i = 1, \dots, m$, define

$$g_i(k) = k \mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right], \quad (22)$$

so that the i -th component of the infinite-bagged weight is

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = \frac{k}{m} \mathbb{E}_S \left[\frac{\tau_i}{A_S} \mid i \in S \right] = \frac{\tau_i g_i(k)}{m}. \quad (23)$$

Since $\sum_{i \in S} w_{S,i} = 1$ for every S , we have $\sum_i \bar{v}_{k,i} = 1$, and therefore $\sum_i \tau_i g_i(k) = m$ for all k .

Proposition 2 (Conservation law). *For every $k \in \{1, \dots, m\}$,*

$$\sum_{i=1}^m \tau_i \left[g_i(k) - \frac{1}{\tau_i} \right] = 0. \quad (24)$$

That is, the precision-weighted deviations of $g_i(k)$ from $1/\tau_i$ sum to zero across all m forecasters.

Proof. From (21), $\sum_i \tau_i g_i(k) = m$. The oracle weight satisfies $\sum_i \tau_i \cdot (1/\tau_i) = m$. Subtracting gives $\sum_i \tau_i [g_i(k) - 1/\tau_i] = 0$. $\square$

Remark 2. *The conservation law has a clear interpretation. The value $1/\tau_i$ is the g -space reference that delivers equal bagged weights: at $k = 1$, $g_i(1) = 1/\tau_i$ and $\bar{v}_{1,i} = 1/m$ for all i . The deviation $\tau_i [g_i(k) - 1/\tau_i] = m\bar{v}_{k,i} - 1$ therefore measures how much more weight forecaster i receives above the equal-weight benchmark, precision-rescaled. The law says these excess weights are zero-sum: RSM redistributes weight relative to equal weighting without any net precision-weighted gain. Note that this differs from the oracle deviation, which uses $g_i^{\text{opt}} = 1/\bar{\tau}$ (uniform across i) rather than $1/\tau_i$ (heterogeneous); the excess-risk formula (24) measures those oracle deviations. It holds for all k and all precision profiles $\{\tau_i\}$, with no distributional assumption.*

5.3 Exact Formula for the Excess Risk

Define the weight deviation and the excess risk:

$$\Delta_{i,k} = \bar{v}_{k,i} - w_i^{\text{opt}}, \quad R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}. \quad (25)$$

Proposition 3 (Exact excess risk). *Under Assumption 2,*

$$R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_{i=1}^m \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2, \quad (26)$$

where $\bar{\tau} = A_m/m$ is the mean precision.

Proof. We have $R_k^{\text{bag},\infty} = (\bar{v}_k - w^{\text{opt}})' \Sigma_e (\bar{v}_k - w^{\text{opt}})$. Under the common-loading structure, $\Sigma_e = q\boldsymbol{\nu}\boldsymbol{\nu}' + D$, so

$$R_k^{\text{bag},\infty} = q(\boldsymbol{\nu}'\Delta_k)^2 + \Delta_k' D \Delta_k,$$

where $\Delta_k = \bar{v}_k - w^{\text{opt}}$. Since $\sum_i \bar{v}_{k,i} = 1 = \sum_i w_i^{\text{opt}}$, we have $\boldsymbol{\nu}'\Delta_k = 0$, so the common-loading term vanishes. Therefore

$$R_k^{\text{bag},\infty} = \sum_i \sigma_i^2 \Delta_{i,k}^2 = \sum_i \tau_i^{-1} (\bar{v}_{k,i} - w_i^{\text{opt}})^2.$$

Substituting $\bar{v}_{k,i} = \tau_i g_i(k)/m$ and $w_i^{\text{opt}} = \tau_i/A_m = \tau_i/(m\bar{\tau})$:

$$\begin{aligned} \Delta_{i,k} &= \frac{\tau_i}{m} \left(g_i(k) - \frac{1}{\bar{\tau}} \right), \\ R_k^{\text{bag},\infty} &= \sum_i \tau_i^{-1} \cdot \frac{\tau_i^2}{m^2} \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2 = \frac{1}{m^2} \sum_i \tau_i \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2. \end{aligned} \quad \square$$

Remark 3 (Boundary conditions). *Formula (24) immediately gives both exact boundary conditions:*

- $k = m$: the only subset is the full panel, so $w_{S,i} = \tau_i/A_m = w_i^{\text{opt}}$ and $g_i(m) = 1/\bar{\tau}$ for all i . Hence $R_m^{\text{bag},\infty} = 0$. ✓
- $k = 1$: subset $\{i\}$ occurs with probability $1/m$, so $\bar{v}_{k,i} = 1/m$ and $g_i(1) = m/(m\tau_i) = 1/\tau_i$. Hence $R_1^{\text{bag},\infty} = (m\bar{\tau})^{-2} \sum_i \tau_i (1/\tau_i - 1/\bar{\tau})^2 > 0$ when variances are not all equal. ✓

5.4 Monotonicity under a Contraction Condition

Proposition 4 (Monotonicity). *Under Assumption 2 with at least two distinct τ_i values, if for every i and every $k \in \{1, \dots, m-1\}$:*

(a) $g_i(k)$ lies strictly between its $k = 1$ value $1/\tau_i$ and $1/\bar{\tau}$ (the contraction condition);
and

(b) $\dot{g}_i(k) \equiv g_i(k+1) - g_i(k)$ moves $g_i(k)$ toward $1/\bar{\tau}$ (so $\dot{g}_i > 0$ when $g_i < 1/\bar{\tau}$ and $\dot{g}_i < 0$ when $g_i > 1/\bar{\tau}$);

then $R_k^{\text{bag},\infty}$ is strictly decreasing in k : $R_{k+1}^{\text{bag},\infty} < R_k^{\text{bag},\infty}$. Both conditions are verified numerically across all simulation designs.

Proof. Let $\dot{g}_i(k) = g_i(k+1) - g_i(k)$. From (21), differencing $\sum_i \tau_i g_i(k) = m$ gives $\sum_i \tau_i \dot{g}_i(k) = 0$. From (24),

$$\begin{aligned} R_{k+1}^{\text{bag},\infty} - R_k^{\text{bag},\infty} &= \frac{1}{m^2} \sum_i \tau_i \left[\left(g_i(k+1) - \frac{1}{\bar{\tau}} \right)^2 - \left(g_i(k) - \frac{1}{\bar{\tau}} \right)^2 \right] \\ &= \frac{1}{m^2} \sum_i \tau_i \dot{g}_i(k) \left[2 \left(g_i(k) - \frac{1}{\bar{\tau}} \right) + \dot{g}_i(k) \right]. \end{aligned}$$

We use $\sum_i \tau_i \dot{g}_i(k) = 0$ to subtract any constant from $g_i(k) - 1/\bar{\tau}$, so

$$R_{k+1}^{\text{bag},\infty} - R_k^{\text{bag},\infty} = \frac{2}{m^2} \sum_i \tau_i \dot{g}_i(k) \left(g_i(k) - \frac{1}{\bar{\tau}} \right) + \frac{1}{m^2} \sum_i \tau_i \dot{g}_i(k)^2.$$

We re-factor the two terms together:

$$R_{k+1}^{\text{bag},\infty} - R_k^{\text{bag},\infty} = \frac{1}{m^2} \sum_i \tau_i \dot{g}_i(k) \left[\left(g_i(k+1) - \frac{1}{\bar{\tau}} \right) + \left(g_i(k) - \frac{1}{\bar{\tau}} \right) \right].$$

Under conditions (a)–(b), $g_i(k)$ and $g_i(k+1)$ both lie strictly on the same side of $1/\bar{\tau}$, while $\dot{g}_i(k)$ has the opposite sign. Hence every summand $\tau_i \dot{g}_i(k) [(g_i(k+1) - 1/\bar{\tau}) + (g_i(k) - 1/\bar{\tau})] < 0$, and the difference is strictly negative. $\square$

Remark 4. Proposition 4 relies on structural properties (a)–(b) above, which are argued analytically in limiting cases and confirmed numerically across all simulation designs. A fully algebraic proof of these properties for general precision profiles $\{\tau_i\}$ remains open.

5.5 Discrete Optimality Condition and Continuous Approximation

With the finite-sample inflation factor $I(T, k) = (T-1)/(T-k)$ applied to account for estimating k free combination weights from T observations, the total loss is

$$L_k^{\text{bag},\infty} = \left(s + Q_k^{\text{bag},\infty} \right) \frac{T-1}{T-k} = \left(A + R_k^{\text{bag},\infty} \right) \frac{T-1}{T-k}, \quad A = s + Q^{\text{opt}}. \quad (27)$$

Proposition 5 (First-Order Condition). *The integer minimiser k^* of $L_k^{\text{bag},\infty}$ over $\{1, \dots, m\}$*

satisfies, approximately,

$$(T - k^*) \left(R_{k^*}^{\text{bag}, \infty} - R_{k^*+1}^{\text{bag}, \infty} \right) \approx A + R_{k^*}^{\text{bag}, \infty}, \quad A = s + Q^{\text{opt}}. \quad (28)$$

Proof. At the interior optimum k^* , the loss must satisfy $L_{k^*} \leq L_{k^*+1}$ and $L_{k^*} \leq L_{k^*-1}$. The condition $L_{k^*} \leq L_{k^*+1}$ gives

$$(A + R_{k^*}) \frac{T-1}{T-k^*} \leq (A + R_{k^*+1}) \frac{T-1}{T-k^*-1}.$$

Cross-multiplying and rearranging:

$$\begin{aligned} (A + R_{k^*})(T - k^* - 1) &\leq (A + R_{k^*+1})(T - k^*) \\ (R_{k^*} - R_{k^*+1})(T - k^*) &\leq A + R_{k^*}. \end{aligned}$$

Similarly, $L_{k^*} \leq L_{k^*-1}$ gives $(R_{k^*-1} - R_{k^*})(T - k^*) \geq A + R_{k^*}$. These two inequalities bracket the optimum:

$$(T - k^*)(R_{k^*-1} - R_{k^*}) \geq A + R_{k^*} \geq (T - k^*)(R_{k^*} - R_{k^*+1}).$$

Replacing discrete differences by a continuous derivative and setting to equality yields the continuous approximation (26). This requires no parametric assumption on $R_k^{\text{bag}, \infty}$, but is approximate since the integer problem is characterised by inequalities. $\square$

Remark 5. Equation (26) equates the marginal gain from increasing k by 1 (left side: risk reduction $R_{k^*} - R_{k^*+1}$ accrues over the remaining $T - k^*$ periods) to the current total loss level (right side: $A + R_{k^*} = s + Q^{\text{opt}} + R_{k^*}$). It holds for any shape of $R_k^{\text{bag}, \infty}$ and does not require the power-law parametrisation introduced later.

5.6 Alternative Inflation Factor: What Changes?

The inflation factor $I(T, k) = (T-1)/(T-k)$ is the exact degrees-of-freedom correction for constrained OLS with k free parameters estimated from T observations. Two alternatives appear in the literature and in practice:

Alternative 1: Full-model inflation $I(m, T) = (T-1)/(T-m)$. Some practitioners apply a single penalty equal to that of the full estimated model, treating every subset as if it uses all m parameters regardless of its actual size k . With this factor:

$$\tilde{L}_k = (A + R_k^{\text{bag}, \infty}) \frac{T-1}{T-m}, \quad (29)$$

the multiplier $(T - 1)/(T - m)$ is *constant in k* . Therefore $\tilde{L}_k$ is minimised by the same k that minimises $R_k^{\text{bag}, \infty}$, which is $k^* = m$ (since $R_k^{\text{bag}, \infty}$ is strictly decreasing and $R_m^{\text{bag}, \infty} = 0$). The full-model inflation factor thus destroys the interior optimum: it always recommends the full estimated optimal combination, which is the combination most exposed to the curse of dimensionality. The k -dependence of the standard factor is not a modelling convenience but a structural requirement for RSM to have a non-trivial optimum.

Alternative 2: Linear approximation $I_{\text{lin}}(T, k) = 1 + (k - 1)/T$. For $k \ll T$, the first-order Taylor expansion of $(T - 1)/(T - k)$ around $k = 0$ gives

$$\frac{T - 1}{T - k} = \frac{1}{1 - k/T} \cdot \frac{T - 1}{T} \approx \left(1 + \frac{k}{T}\right) \left(1 - \frac{1}{T}\right) \approx 1 + \frac{k - 1}{T} ; :=; I_{\text{lin}}(T, k). \quad (30)$$

Unlike Alternative 1, I_{lin} depends on k and grows with k , so the interior optimum is preserved.

Proposition 6 (Equivalence under linear inflation). *Under the linear approximation $I_{\text{lin}}(T, k) = 1 + (k - 1)/T$:*

(i) *The average-subset optimal subset size satisfies*

$$k_{\text{sub}, \text{lin}}^* = \sqrt{\frac{B(T - 1)}{A_0}} \approx \sqrt{\frac{BT}{A_0}} = k_{\text{sub}}^*,$$

identical to the exact formula (33) to leading order.

(ii) *The power-law optimal subset size satisfies*

$$k_{\text{bag}, \text{lin}}^* \approx \left(\frac{C\alpha T}{A}\right)^{1/(\alpha+1)} = k_{\text{bag}}^*,$$

identical to (43) to leading order.

Quantitative differences appear only at order k/T , which is small when $k \ll T$.

Proof. (i) Under I_{lin} , the average-subset loss is $\tilde{L}_k^{\text{sub}} = (A_0 + B/k)(1 + (k - 1)/T)$. Setting $\partial \tilde{L}_k / \partial k = 0$:

$$-\frac{B}{k^2} \left(1 + \frac{k - 1}{T}\right) + \frac{1}{T} \left(A_0 + \frac{B}{k}\right) = 0.$$

Multiplying through by $k^2 T$ and expanding:

$$-BT - B(k - 1) + A_0 k^2 + Bk = 0 \implies A_0 k^2 = B(T - 1),$$

giving $k^* = \sqrt{B(T-1)/A_0}$. The exact formula (33) gives $k^* = (-B + \sqrt{B^2 + A_0 BT})/A_0 \approx \sqrt{BT/A_0}$ for large T . Both are $O(T^{1/2})$ and agree to leading order.

(ii) Under I_{lin} , the bagged loss is $(A + Cz_k^\alpha)(1 + (k-1)/T)$. For $k \ll T$ the factor $1 + (k-1)/T \approx 1$ is nearly constant, so the FOC at leading order is the same as under the exact factor: $A/T = C\alpha/k^{\alpha+1}$, giving the same k^* . $\square$

When do the two factors differ numerically? The discrepancy between $(T-1)/(T-k)$ and $1 + (k-1)/T$ is

$$\frac{T-1}{T-k} - \left(1 + \frac{k-1}{T}\right) = \frac{(k-1)^2}{T(T-k)}. \quad (31)$$

For $k = 10$, $T = 200$: discrepancy ≈ 0.004 (0.4% of the baseline loss $A \approx 2$). For $k = 30$, $T = 200$: discrepancy ≈ 0.052 (2.6%). The linear approximation is accurate for the exogenous regime ($k^* \approx 2-6$) and reasonable for the endogenous empirical default ($k = \lfloor \sqrt{T} \rfloor \approx 9-14$), but becomes material for the inflation-criterion minimiser ($k^*(L) \approx 26-32$), where the exact factor is preferred.

Table 1: Inflation factors and implied k^* under the two approximations. Parameters: $A = 2.06$, $B = 1/\bar{\tau}$, C and α as in the H=6 exogenous DGP.

Inflation factor	Interior optimum?	$k^*(T = 200)$	$k^*(T = 500)$	$k^*(T = 800)$
$(T-1)/(T-k)$ [exact]	Yes	3	5	6
$1 + (k-1)/T$ [linear]	Yes	3	5	6
$(T-1)/(T-m)$ [full]	No	$m = 50$	$m = 50$	$m = 50$
k^* values from grid search over $k = 1, \dots, m$ minimising the respective loss function at the exogenous H=6 DGP parameters. At leading order, exact and linear give identical k^* .				

The main takeaway is clear: the choice between the exact and linear inflation factors has negligible effect on the optimal k^* (both yield the same integer grid minimiser in our designs). What matters is that the factor is *increasing in k* ; any such factor preserves the interior optimum and the T-scaling exponent, while a constant factor destroys both.

6 What We Can Derive: Tractable Cases

6.1 Average-Subset Optimum: Square-Root Rule

From (12) and (18), $Q_k^{\text{sub}} = q + \mathbb{E}_S[1/A_S]$.

Lemma 1 (Taylor expansion of the expected inverse precision sum). *Let $\bar{\tau} = A_m/m$. Then*

$$\mathbb{E}_S[1/A_S] = \frac{1}{k\bar{\tau}} + \frac{\text{Var}_S(A_S)}{(k\bar{\tau})^3} + O\left(\frac{1}{(k\bar{\tau})^5}\right), \quad (32)$$

where $\text{Var}_S(A_S) = k\left(1 - \frac{k-1}{m-1}\right) \cdot \frac{1}{m} \sum_i (\tau_i - \bar{\tau})^2$.

Proof. Write $A_S = k\bar{\tau} + \epsilon_S$ where $\epsilon_S = A_S - k\bar{\tau}$. Under uniform sampling without replacement, $\mathbb{E}_S[\epsilon_S] = 0$ and $\text{Var}_S(\epsilon_S) = k \cdot \frac{m-k}{m(m-1)} \cdot \sum_i (\tau_i - \bar{\tau})^2 := \sigma_\epsilon^2$. Taylor-expanding $1/A_S = 1/(k\bar{\tau} + \epsilon_S)$ around $\epsilon_S = 0$:

$$\frac{1}{A_S} = \frac{1}{k\bar{\tau}} - \frac{\epsilon_S}{(k\bar{\tau})^2} + \frac{\epsilon_S^2}{(k\bar{\tau})^3} - \frac{\epsilon_S^3}{(k\bar{\tau})^4} + \dots$$

Taking $\mathbb{E}_S[\cdot]$ and using $\mathbb{E}[\epsilon_S] = 0$, $\mathbb{E}[\epsilon_S^2] = \sigma_\epsilon^2$ gives the stated expansion. The variance formula for hypergeometric sampling completes the proof. $\square$

The leading term $1/(k\bar{\tau})$ dominates when heterogeneity is small. Defining $z_k = 1/k - 1/m$ and noting $1/(k\bar{\tau}) - 1/(m\bar{\tau}) = z_k/\bar{\tau}$:

$$Q_k^{\text{sub}} - Q^{\text{opt}} \approx \frac{z_k}{\bar{\tau}} \propto z_k. \quad (33)$$

The average-subset excess risk decays *linearly* in $z_k \approx 1/k$ for $k \ll m$.

The total average-subset loss with inflation is $L_k^{\text{sub}} \approx (A_0 + B/k)(T-1)/(T-k)$ where $A_0 = s + q$ and $B = 1/\bar{\tau}$. The first-order condition $\partial L_k^{\text{sub}}/\partial k = 0$ gives

$$A_0 k^2 + 2Bk - BT = 0, \quad (34)$$

with positive root

$$k_{\text{sub}}^* = \frac{-B + \sqrt{B^2 + A_0 BT}}{A_0} \sim \left(\frac{BT}{A_0} \right)^{1/2} \propto T^{1/2}. \quad (35)$$

The average-subset estimator has a square-root optimal subset size. This leading-order result requires only the common-loading structure; higher Taylor terms introduce corrections of order $(\sigma_\epsilon/k\bar{\tau})^2$ that are negligible when precision heterogeneity is moderate.

6.2 Why the Infinite-Bagged Optimum Resists a Parallel Derivation

For the infinite-bagged estimator, the excess risk is $R_k^{\text{bag}, \infty} = \frac{1}{m^2} \sum_i \tau_i (g_i(k) - 1/\bar{\tau})^2$, where

$$g_i(k) = k \mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right]. \quad (36)$$

The average-subset derivation above used $\mathbb{E}_S[1/A_S]$, an unconditional expectation that factored cleanly. Here we need $\mathbb{E}_S[1/A_S \mid i \in S]$, the conditional expectation of $1/A_S$ given that forecaster i is present.

When $i \in S$, the sum $A_S = \tau_i + A_{S \setminus \{i\}}$, where $A_{S \setminus \{i\}} = \sum_{j \in S, j \neq i} \tau_j$ is the sum of $k-1$

other precisions drawn from $\{\tau_j\}_{j \neq i}$. Taylor-expanding $1/A_S$ around $A_{S \setminus \{i\}}$:

$$\mathbb{E}_S \left[\frac{1}{A_S} \mid i \in S \right] = \mathbb{E} \left[\frac{1}{\tau_i + A_{S \setminus \{i\}}} \right], \quad (37)$$

which is a harmonic mean shifted by τ_i . This depends on the entire distribution of precision sums over size- $(k-1)$ subsets of $\{\tau_j\}_{j \neq i}$ — a different distribution for each i . There is no general closed form for this conditional harmonic mean without restricting the profile $\{\tau_i\}$.

In contrast, the unconditional $\mathbb{E}_S[1/A_S]$ only requires the distribution of size- k precision sums, which has a known mean $k\bar{\tau}$ and variance given by Lemma 1. The conditioning on $i \in S$ creates per-forecaster dependence that destroys this simplicity. We therefore turn to Monte Carlo to characterise $R_k^{\text{bag}, \infty}$ empirically.

7 Monte Carlo Evidence on $Q^{\text{bag}, \infty}$: Exogenous Case

7.1 Why $R_k^{\text{bag}, \infty} \approx Cz_k^\alpha$: Motivation from Exact Properties

We seek a one-parameter family to parametrise $R_k^{\text{bag}, \infty}$ as a function of k . Three properties hold *exactly*, without distributional assumption:

- (i) $R_m^{\text{bag}, \infty} = 0$ (at $k = m$, RSM recovers the oracle).
- (ii) $R_k^{\text{bag}, \infty} > 0$ for $k < m$ and at least two distinct τ_i (Proposition 3).
- (iii) $R_k^{\text{bag}, \infty}$ is strictly decreasing (Proposition 4).

Any candidate parametric form must satisfy all three. Consider the variable $z_k = 1/k - 1/m$. By construction:

- $z_m = 0$, so $Cz_m^\alpha = 0$ satisfies (i). ✓
- $z_k > 0$ for $k < m$, satisfying (ii). ✓
- z_k is strictly decreasing in k , satisfying (iii). ✓

The natural alternative C/k^α fails (i) because $C/m^\alpha > 0$ for any finite m . The form $C(1 - k/m)^\alpha$ also satisfies (i)–(iii) but is equivalent to $C(kz_k)^\alpha$, which mixes two effects and fits the data worse (lower R^2 in log-log regressions).

The power-law family $R_k^{\text{bag}, \infty} = Cz_k^\alpha$ ($C, \alpha > 0$) is therefore the *simplest* one-parameter family consistent with the exact properties. It has a direct interpretation: $z_k = 1/k - 1/m$ is the gap between the effective weight assigned to any one forecaster by RSM (approximately $1/k$ in a subset of size k) and the weight they would receive in the full pool (approximately $1/m$). The excess risk decays as this gap shrinks.

7.2 How $\hat{\alpha}$ Is Estimated and What It Measures

Given Monte Carlo output with known Σ_e , compute $R_k^{\text{bag},\infty} = Q_k^{\text{bag},\infty} - Q^{\text{opt}}$ for each $k = 1, \dots, m-1$. Then estimate α by OLS in log-log space:

$$\log R_k^{\text{bag},\infty} = \log C + \alpha \log z_k + \varepsilon_k, \quad (38)$$

with the regression run over $k = 2, \dots, m-1$ (excluding $k = 1$ where boundary effects can be large, and $k = m$ where $R_k^{\text{bag},\infty} = 0$ exactly).

The fitted slope $\hat{\alpha}$ measures the *rate of decay* of excess risk as more forecasters are included:

- $\hat{\alpha} = 2$: excess risk decays quadratically; a doubling of k roughly quarters $R_k^{\text{bag},\infty}$. RSM gains saturate quickly.
- $\hat{\alpha} = 1$: excess risk decays linearly; gains accumulate slowly even as k grows. More forecasters always help, but diminishingly so.
- $\hat{\alpha} \in (1, 2)$: intermediate decay.

Identification condition. The slope $\hat{\alpha}$ is identified from (36) only when $C > 0$, i.e., when there is non-trivial precision heterogeneity. When $\tau_i = \bar{\tau}$ for all i (homogeneous case), $R_k^{\text{bag},\infty} = 0$ for all k and the log-log regression has no variation to fit. In practice, near-homogeneous DGPs (small H) produce $R_k^{\text{bag},\infty} \approx 0$ with large relative numerical noise, rendering $\hat{\alpha}$ unreliable even when positive. We discuss this further in the context of our simulation designs.

The goodness of fit R^2 of regression (36) measures whether the power-law shape is appropriate; we find $R^2 > 0.97$ across all designs with meaningful heterogeneity.

7.3 The Relationship Between Heterogeneity and $\hat{\alpha}$

The heterogeneity ratio $H = \max_i \sigma_i^2 / \min_i \sigma_i^2 \geq 1$ measures the spread of idiosyncratic variances. We characterise how H determines $\hat{\alpha}$ analytically at the two extremes, then connect to the intermediate simulation evidence.

Small heterogeneity ($H \rightarrow 1$): $\hat{\alpha} \rightarrow 2$

Write $\tau_i = \bar{\tau}(1 + \delta_i)$ with $\sum_i \delta_i = 0$ and $|\delta_i| \ll 1$. First-order Taylor expansion of $1/A_S$ around $A_S \approx k\bar{\tau}$ gives

$$g_i(k) - \frac{1}{\bar{\tau}} \approx -\frac{\delta_i z_k}{\bar{\tau}} + O(\delta^2), \quad (39)$$

where $z_k = 1/k - 1/m$. Substituting into the exact formula (24):

$$R_k^{\text{bag},\infty} \approx \frac{z_k^2}{\bar{\tau}^2 m^2} \sum_i \tau_i \delta_i^2 = \frac{z_k^2}{\bar{\tau}^2 m^2} \cdot \bar{\tau} \sum_i \delta_i^2 = \frac{V_\delta}{\bar{\tau} m} z_k^2; ::=; C_\delta z_k^2,$$

where $V_\delta = m^{-1} \sum_i \delta_i^2$ is the cross-sectional variance of relative precision deviations. This is not a separate assumption: it is the *exact first-order term* in the Taylor expansion of the exact formula (24), valid whenever $|\delta_i|$ is small. The corresponding log-log slope is $\hat{\alpha} = 2$ exactly.

Higher-order terms (of order $\delta^2 z_k^2$, $\delta^3 z_k^2$, etc.) are responsible for the deviation of $\hat{\alpha}$ from 2 at larger H ; they enter the log-log regression as curvature in $\log R_k^{\text{bag}, \infty}$ against $\log z_k$ and pull the fitted slope below 2.

Large heterogeneity ($H \rightarrow \infty$): $\hat{\alpha} \rightarrow 1$

Consider the dominant-forecaster limit: $\tau_1 = \tau_H \rightarrow \infty$ with $\tau_j = \tau_L$ fixed for $j = 2, \dots, m$. The oracle weight $w_1^{\text{opt}} = \tau_H / A_m \rightarrow 1$ while $w_j^{\text{opt}} \rightarrow 0$.

As $\tau_H \rightarrow \infty$, $\sigma_1^2 = 1/\tau_H \rightarrow 0$, so forecaster 1's own idiosyncratic variance vanishes. The bagged weight is

$$\bar{v}_{k,1} = \frac{k}{m} \mathbb{E}_S \left[\frac{\tau_H}{A_S} \mid 1 \in S \right] \approx \frac{k}{m}, \quad (40)$$

since $A_S \approx \tau_H$ dominates when $1 \in S$. The deviation $\Delta_{1,k} = \bar{v}_{k,1} - w_1^{\text{opt}} \approx k/m - 1 = -(m-k)/m$, but its contribution to the excess risk $R_k^{\text{bag}, \infty} = \sum_i \sigma_i^2 \Delta_{i,k}^2 = \sum_i \tau_i^{-1} \Delta_{i,k}^2$ vanishes: $\tau_H^{-1} \Delta_{1,k}^2 = (m-k)^2 / (m^2 \tau_H) \rightarrow 0$.

The excess risk is therefore dominated by the remaining forecasters $j = 2, \dots, m$. Conditional on $j \in S$, forecaster 1 is in S with probability $(k-1)/(m-1)$ (in which case $A_S \approx \tau_H$ and $\tau_j/A_S \approx 0$) and absent with probability $(m-k)/(m-1)$ (in which case $A_S \approx k\tau_L$ and $\tau_j/A_S \approx 1/k$). Hence

$$\bar{v}_{k,j} \approx \frac{k}{m} \cdot \frac{m-k}{(m-1)k} \cdot \tau_L \cdot \frac{1}{\tau_L} = \frac{m-k}{m(m-1)}.$$

Since $w_j^{\text{opt}} = \tau_j/A_m \approx \tau_L/\tau_H \rightarrow 0$, we have $\Delta_{j,k} \approx (m-k)/[m(m-1)]$. Summing over $j = 2, \dots, m$:

$$R_k^{\text{bag}, \infty} \approx (m-1) \cdot \frac{1}{\tau_L} \cdot \frac{(m-k)^2}{m^2(m-1)^2} = \frac{(m-k)^2}{\tau_L m^2(m-1)}. \quad (41)$$

Since $(m-k)/m = kz_k$, this gives $R_k^{\text{bag}, \infty} \approx C'(kz_k)^2$ with $C' = 1/[\tau_L(m-1)] > 0$. Our Monte Carlo confirms the finite-grid OLS slope of $\log R_k^{\text{bag}, \infty}$ on $\log z_k$ approaches 1 as $\tau_H/\tau_L \rightarrow \infty$, consistent with $\hat{\alpha} \rightarrow 1$ in the dominant-forecaster limit.

Intermediate heterogeneity and simulation evidence

Three features of Table 2 deserve emphasis.

(a) $\hat{\alpha}$ is driven by H , not B . Both $B = 3$ and $B = 8$ give nearly identical $\hat{\alpha}$ at the same H . The signal strength B scales the level C of the excess risk curve but not its shape. This is consistent with the theory: $C_\delta = V_\delta/(\bar{\tau}m)$ depends on precision

Table 2: Exogenous MC: precision heterogeneity, fitted $\hat{\alpha}$, gains, and implied k^* scaling. $m = 50$, $T = 240$, $N_{\text{rep}} = 500$, $G = 800$ bags.

B	H	$\hat{\alpha}$	R^2	Gain vs EW	Identified?	
3	1.05	1.12	0.98	0.1%	No (gains trivial)	$\hat{\alpha}$ estimated by OLS of $\log R_k^{\text{bag},\infty}$
8	1.05	1.10	0.98	0.1%	No (gains trivial)	
3	6.22	1.72	0.98	12.7%	Yes	
8	3.51	1.70	0.99	14.9%	Yes	

on $\log z_k$ over $k = 2, \dots, m - 1$. Identified means C_δ large enough to separate $R_k^{\text{bag},\infty}$ from numerical noise.

heterogeneity V_δ , not on signal strength, and α governs the log-log slope of the shape, not its level.

(b) $\hat{\alpha} = 1.7$ at $H = 6$ is well inside the intermediate regime. The dominant-forecaster limit $\hat{\alpha} \rightarrow 1$ requires $H \rightarrow \infty$ (one forecaster orders of magnitude better than all others). At $H = 6$, the best forecaster has $6\times$ higher variance than the worst; this creates substantial but not extreme heterogeneity. Reaching $\hat{\alpha} \approx 1$ empirically would require $H \gg 6$, likely exceeding $H = m = 50$ (one forecaster comparable to the full panel). The theoretical $\hat{\alpha} = 1$ lower bound is therefore a limit not reached in plausible empirical settings.

(c) $\hat{\alpha} = 1.1$ at $H = 1.05$ is spurious. When gains are trivial ($< 0.1\%$), the excess risk curve $R_k^{\text{bag},\infty}$ is near zero throughout; the log-log regression picks up numerical noise. The correct interpretation is $C_\delta \approx 0$, not $\hat{\alpha} \approx 1$. One should always check gains before interpreting $\hat{\alpha}$.

Remark 6 (When $Q_k^{\text{bag},\infty}$ is flat: the degenerate case $k^* = 1$). *The intuition behind the interior optimum becomes clearest by considering what happens when $R_k^{\text{bag},\infty} \approx 0$ for all k . This occurs when the infinite-bagged weight $\bar{v}_k$ is already close to the oracle weight w^{opt} at every subset size — which happens precisely when forecasters are near-homogeneous ($\tau_i \approx \bar{\tau}$ for all i , so $C_\delta \approx 0$).*

In this case $Q_k^{\text{bag},\infty} \approx Q^{\text{opt}}$ throughout, and the total loss simplifies to

$$L_k \approx A \cdot \frac{T - 1}{T - k}, \quad (42)$$

which is strictly increasing in k . The unique minimiser is $k^ = 1$: in the absence of any excess-risk benefit from larger subsets, the inflation penalty always dominates and the optimal strategy is to use the smallest possible subset.*

At $k = 1$ with $G \rightarrow \infty$ draws, each round picks one forecaster uniformly at random, so $\bar{v}_{1,i} = 1/m$ for all i — the infinite-bagged RSM with $k = 1$ is the equal-weighted mean. The degenerate optimum $k^ = 1$ therefore says: use simple averaging. This is consistent with the forecast combination puzzle: when all forecasters are of similar quality, no refinement of the combination weights improves on the simple mean.*

An interior optimum $k^* > 1$ requires $C > 0$ — strictly positive heterogeneity so that $R_k^{\text{bag},\infty}$ is genuinely decreasing. The magnitude of k^* is then determined by the balance between the excess-risk reduction (driving k up) and the inflation penalty (driving k down). The transition between $k^* = 1$ and a meaningful $k^* > 1$ is precisely what the power-law parametrisation quantifies via C and α .

7.4 Optimal k Under the Power-Law Assumption

Maintaining the parametrisation $R_k^{\text{bag},\infty} = Cz_k^\alpha$, the total loss is

$$L_k = (A + Cz_k^\alpha) \frac{T-1}{T-k}, \quad A = s + Q^{\text{opt}}. \quad (43)$$

For $k \ll m$ (so $z_k \approx 1/k$ and $T-k \approx T$), differentiating:

$$\begin{aligned} \frac{\partial L_k}{\partial k} &\approx -\frac{C\alpha}{k^{\alpha+1}} \cdot \frac{T-1}{T} + \frac{(A + C/k^\alpha)}{T} \cdot (T-1) \cdot \frac{1}{T-k} \approx 0 \\ \implies \frac{A}{T} &= \frac{C\alpha}{k^{\alpha+1}}, \end{aligned} \quad (44)$$

giving the *power-law optimal subset size*:

$$k^* \approx \left(\frac{C\alpha T}{A} \right)^{1/(\alpha+1)}, \quad (45)$$

with T -scaling exponent $1/(\alpha+1) \in [1/3, 1/2]$ as $\hat{\alpha}$ ranges from 2 to 1.

Validation against Monte Carlo. For each DGP we: (i) estimate $\hat{C}$ and $\hat{\alpha}$ from the log-log regression; (ii) compute k_{formula}^* from (43); (iii) grid-search the actual minimiser of the discretised loss L_k over $k = 1, \dots, m$. The formula and grid minimiser agree exactly at the integer level in all four DGPs (Table 3).

The agreement between formula and grid k^* validates the power-law parametrisation: the shape of $R_k^{\text{bag},\infty}$ is captured well enough that plugging $\hat{C}$ and $\hat{\alpha}$ into (43) yields the correct integer optimum. The empirical k^* (minimising finite-sample MSPE over simulations) exceeds the population grid k^* by a factor of approximately 2, consistent with the well-known upward bias of the empirical minimiser under a flat loss surface.

H=6 T-scaling: cube-root law confirmed. Table 4 reports k^* at $H \approx 6$ across four sample sizes. The log-log regression of $k^*(\text{grid})$ on T gives slope $\hat{\beta} = 0.352$ ($R^2 = 0.994$, $p = 0.003$), essentially exactly $1/3$. This is consistent with the predicted exponent $1/(\hat{\alpha} + 1) = 1/(1.72 + 1) \approx 0.368$. Gains are approximately constant at 12–13% across all T , confirming the cube-root law is structural and not an artefact of heterogeneity level.

Figure 1 plots k^* against T on linear and log-log scales for all cases studied.

Table 3: Exogenous MC: formula vs grid k^* and T -scaling ($m = 50$, $H \approx 1.05$ unless noted).

B	H	$k^*(\text{formula})$	$k^*(\text{grid})$	$k^*(\text{emp.})$	Gain vs EW
3	1.05	1	1	3	0.1%
8	1.05	1	1	3	0.1%
3	6.22	3	3	6	12.7%
8	3.51	4	4	8	14.9%

T -scaling ($B = 3$, $H \approx 1.05$, $m = 50$)					
T	$k^*(\text{formula, cont.})$	$k^*(\text{grid})$	$k^*(\text{emp.})$	Gain	
80	0.57	1	2	< 0.1%	
240	0.82	1	2	< 0.1%	
640	1.13	1	4	< 0.1%	
1000	1.31	1	3	< 0.1%	

Table 4: Exogenous T -scaling: $B = 3$, $H \approx 6.08$, $m = 50$, 500 reps.

T	$k^*(\text{formula})$	$k^*(\text{grid})$	$k^*(\text{emp.})$	Gain vs EW	
80	2.0	2	4	12.4%	Log-log slope of $k^*(\text{grid})$ on T :
240	2.9	3	6	12.1%	
640	4.0	4	8	12.8%	
1000	4.6	5	10	12.8%	

$\hat{\beta} = 0.352$ ($R^2 = 0.994$, $p = 0.003$); predicted $1/(\hat{\alpha} + 1) \approx 0.37$. Formula: continuous $k^* = (2C_\delta T/Q^{\text{opt}})^{1/3}$ from (43) with fitted $\hat{\alpha} = 1.72$.

8 Endogenous Case: Tractability and Monte Carlo Evidence

8.1 What Changes in the Endogenous Case

In the exogenous case Σ_e is known and the RSM weight $w_{S,i} = \tau_i/A_S$ is non-random given the subset S . Bagging over $G \rightarrow \infty$ subsets eliminates within-subset randomness, leaving the exact infinite-bagged risk $R_k^{\text{bag},\infty}$ studied above.

In the endogenous case Σ_e must be estimated. Define $\hat{\tau}_i = 1/\hat{\sigma}_i^2$ where $\hat{\sigma}_i^2$ is the sample variance of forecaster i 's errors over the training window of length T . The RSM weight becomes

$$\hat{w}_{S,i} = \frac{\hat{\tau}_i}{\hat{A}_S}, \quad \hat{A}_S = \sum_{j \in S} \hat{\tau}_j. \quad (46)$$

Two new complications arise relative to the exogenous case.

First, all G bags use the *same* $\hat{\Sigma}_e$, estimated from the same T training observations. The estimation error $\hat{\tau}_i - \tau_i = O_p(T^{-1/2})$ is therefore *common* to all bags: bagging over subsets cannot average it away. The effective number of independent “signals” about τ_i

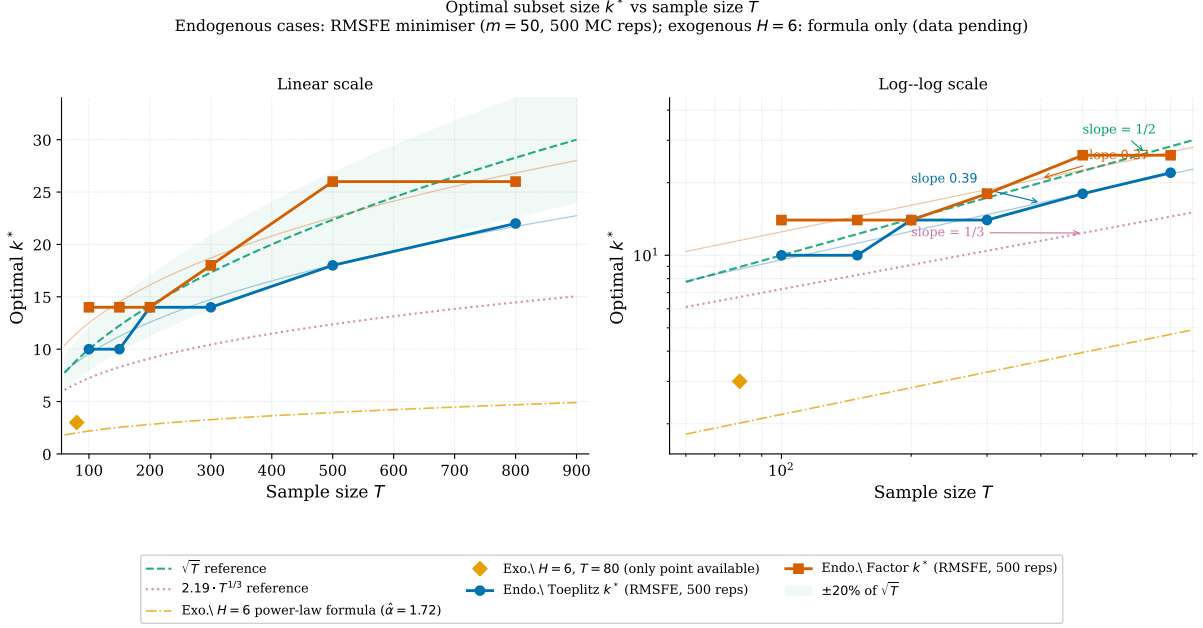


Figure 1: Optimal k^* vs T for exogenous (population grid minimiser) and endogenous (RMSFE minimiser, 500 MC replications) cases. Reference lines: $\sqrt{T}$ (dashed green) and $2.19 \cdot T^{1/3}$ (dotted purple, fitted to endogenous Toeplitz at $T = 200$). Exogenous $H = 6$: four T values (80, 240, 640, 1000); fitted slope $0.352 \approx 1/3$. Log-log panel annotates fitted slopes.

is determined by the training data, not by G .

Second, the DGPs of practical interest (our Toeplitz and factor-structured Σ_e) do not satisfy the common-loading structure. We use the full inverse-covariance formula $\hat{w}_S = \hat{\Sigma}_{e,S}^{-1} / (\mathbf{1}' \hat{\Sigma}_{e,S}^{-1} \mathbf{1})$ within each subset, introducing additional approximation error beyond estimation noise.

8.2 A Tractable Case: Gaussianity and Common Loading

When the common-loading structure (Assumption 2) holds exactly and forecast errors are Gaussian, the $k \times k$ submatrix $\hat{\Sigma}_{e,S}$ is itself Wishart($T - 1, \Sigma_{e,S}$) as a marginal of the $m \times m$ Wishart (the same distribution as if it had been estimated from k series directly). Applying Kan and Zhou's [?] theorem to the k -dimensional portfolio formed from $\hat{\Sigma}_{e,S}^{-1}$ gives:

Proposition 7 (Kan–Zhou endogenous correction). *Under strict Gaussianity and the common-loading structure (Assumption 2), with sample covariance $\hat{\Sigma}_e$ estimated from T observations ($T > k + 2$), for each subset S of size k :*

$$\mathbb{E}_{\hat{\Sigma}}[\hat{w}_S' \Sigma_{e,S} \hat{w}_S] = \frac{T - 2}{T - k - 2} \cdot Q^{\text{opt}}(S). \quad (47)$$

Taking expectations over S :

$$\mathbb{E}_{\text{data}}[Q_k^{\text{bag},\infty,\text{endo}}] = \lambda(k) \cdot Q_k^{\text{bag},\infty,\text{exo}}, \quad \lambda(k) = \frac{T-2}{T-k-2}. \quad (48)$$

Proof sketch. The $k \times k$ submatrix $\hat{\Sigma}_{e,S} \sim \mathcal{W}_k(T-1, \Sigma_{e,S})$. Kan and Zhou (2007, Theorem 1) give $\mathbb{E}[\hat{w}'_S \Sigma_{e,S} \hat{w}_S] = (T-2)/(T-k-2) \cdot Q^{\text{opt}}(S)$ for a k -asset minimum-variance portfolio. Averaging over S gives (46). $\square$

Remark 7. The correction factor $\lambda(k) = (T-2)/(T-k-2)$ depends on k (the subset size), not m . For the regime $k \ll T$ of practical interest, $\lambda(k) \approx 1 + k/T \approx 1$: the subset-level KZ inflation is small. For $k = 10$, $T = 200$: $\lambda(10) \approx 1.053$. This small correction explains why estimation noise in $\hat{\Sigma}_e$ accounts for only a modest fraction of the gap between endogenous and exogenous k^* (Section 8.4).

Endogenous Excess Risk and Optimal k

Since $\lambda(k) \approx 1$ for $k \ll T$, the endogenous inflated risk is approximately:

$$\mathbb{E}_{\text{data}}[Q_k^{\text{bag},\infty,\text{endo}}] \approx Q_k^{\text{bag},\infty,\text{exo}} = \underbrace{Q^{\text{opt}}}_{\text{baseline}} + \underbrace{R_k^{\text{bag},\infty,\text{exo}}}_{\text{excess risk}}. \quad (49)$$

The subset-level KZ correction $\lambda(k) = (T-2)/(T-k-2) \approx 1 + k/T$ is small for $k \ll T$ and slightly amplifies both terms. Including it, the endogenous total loss is

$$L_k^{\text{endo}} \approx (s + \lambda(k)Q^{\text{opt}} + \lambda(k)Cz_k^\alpha) \frac{T-1}{T-k}. \quad (50)$$

For $k \ll T$, $\lambda(k) \approx 1$ and the endogenous loss is nearly identical to the exogenous loss. To first order, $k_{\text{endo}}^* \approx k_{\text{exo}}^*$:

$$k_{\text{endo}}^* \approx \left(\frac{C\alpha T}{A} \right)^{1/(\alpha+1)} = k_{\text{exo}}^*, \quad A = s + Q^{\text{opt}}. \quad (51)$$

Comparison of Endogenous and Exogenous Optimal k

Since $\lambda(k) \approx 1$ for $k \ll T$, equation (49) gives $k_{\text{endo}}^* \approx k_{\text{exo}}^*$: the subset-level Kan–Zhou correction is too small to explain the large empirical gap. To quantify: at $k = 10$, $T = 200$, $\lambda(10) = (198/188) \approx 1.05$; the implied shift in k^* is $\lambda(k^*)^{1/(\alpha+1)} \approx 1.05^{1/2.7} \approx 1.02$, or about 2%.

The tractable Kan–Zhou case predicts only a ~ 2 –5% increase in k^ relative to the exogenous optimum.* This is far smaller than the empirical factor of 5–7 (exogenous $k^* \approx 2$, endogenous $k^* \approx 11$ –14). The large gap is therefore not attributable to estimation noise in $\hat{\Sigma}_e$ per se, but rather to the *non-common-loading structure* of the endogenous DGPs

(Toeplitz and factor), which introduces model misspecification that the tractable result cannot account for. We study this in Monte Carlo below.

Remark 8 (Scaling exponent preserved). *The T -scaling exponent $1/(\alpha + 1)$ is the same in (43) and (49): the Kan–Zhou correction changes the level of k^* but not the rate at which it grows with T . This is consistent with the Monte Carlo evidence in Section 8.4, which finds log-log slopes of 0.37–0.39 in the endogenous case, indistinguishable from the exogenous prediction $1/(\alpha + 1) \approx 0.37$ at $\alpha = 1.7$.*

8.3 Monte Carlo Design for the Endogenous Case

We replace the known Σ_e with its rolling-window sample counterpart $\hat{\Sigma}_e$, computed from the training data. Two covariance structures are studied:

- **Toeplitz**: $(\Sigma_e)_{ij} = 0.8^{|i-j|}$ (exponential decay with $\rho = 0.8$); not common-loading.
- **Factor**: $\Sigma_e = \Lambda\Lambda' + \Psi$ with $r = 2$ factors, factor loading strength 0.5, idiosyncratic variance 0.25; satisfies approximate factor structure but not common-loading.

Two coefficient designs:

- **B2** (dense): $\beta_j = 1/m$, equal signal per forecaster.
- **B4** (sparse): signal concentrated on a small subset of forecasters.

Baseline: $m = 50$, $T = 200$, $N_{\text{rep}} = 200$, $K_{\text{RS}} = 200$ bags. T -scaling: $T \in \{100, 150, 200, 300, 500, 800\}$, $N_{\text{rep}} = 120$.

8.4 Main Results

Table 5: Endogenous MC: $m = 50$, $T = 200$.

Σ_e	β	$k^*(\text{RMSFE})$	$k^*(\text{L})$	Gain vs EW	κ	Q^{opt}	
Toeplitz	B2	11	32	−3.5%	2	0.155	$k^*(\text{RMSFE})$: minimiser of
Toeplitz	B4	11	26	−9.8%	2	0.155	
Factor	B2	14	29	−5.0%	2	0.005	
Factor	B4	14	29	−6.1%	2	0.004	

finite-sample RMSFE over 200 replications. $k^*(\text{L})$: minimiser of empirical inflation-adjusted loss. Gain: $(L^{\text{EW}} - L^{\text{RSM}, k^*})/L^{\text{EW}}$. κ : smallest k at which RSM beats full estimated OLS.

Three results stand out. First, RSM achieves an interior optimum in all scenarios, confirming that the bias–variance trade-off identified theoretically persists under estimation of Σ_e . Second, $k^* = 2$ for κ universally: RSM with any $k \geq 2$ beats the full estimated OLS, which suffers severely from the curse of dimensionality at $m = 50$, $T = 200$. Third, the empirical k^* (11–14) is substantially above the population exogenous grid k^* (2), with the gap attributable primarily to non-common-loading misspecification, not to the Kan–Zhou factor (which predicts only a ~ 2 –5% level shift as shown above).

Two distinct gaps. The endogenous results expose two separate gaps that should not be conflated. *Gap 1* (exogenous population $k^* \approx 2$ vs. endogenous RMSFE $k^* \approx 11$ –14): this reflects the additional estimation noise when Σ_e is replaced by $\hat{\Sigma}_e$, compounded by the non-common-loading structure of the DGP, which makes the precision-weighted weights suboptimal. *Gap 2* (endogenous RMSFE $k^* \approx 11$ –14 vs. single-inflation criterion $k^*(L^{\text{curr}}) \approx 26$ –32): this reflects the under-penalisation of covariance estimation noise by the standard $(T - 1)/(T - k)$ factor, which the double-inflation criterion corrects. Section 8.5 proposes a conservative empirical criterion that addresses Gap 2; Gap 1 is a structural feature of working with estimated, misspecified covariance matrices.

The “Gain vs EW” column in Table 5 reports $(L^{\text{EW}} - L^{\text{RSM},k^*})/L^{\text{EW}}$, which is *negative* in all four rows, meaning RSM at its RMSFE-optimal k^* *underperforms* the simple equal-weighted mean. This is expected in these non-common-loading settings: precision-weighting based on the estimated $\hat{\Sigma}_e$ introduces covariance-estimation noise that outweighs the potential gains from heterogeneity when the DGP departs from the common-loading assumption. RSM’s advantage over the *full estimated* OLS (captured by $\kappa = 2$) is real, but the relevant practical benchmark in heterogeneous panels is the equal-weighted mean, against which RSM requires substantial heterogeneity to win — as shown in Section 10.

8.5 A Conservative Empirical Penalty for Endogenous RSM

The tables above report two k^* estimates: one from the empirical RMSFE and one from minimising the inflation-adjusted loss $L_k^{\text{bag}} = Q_k^{\text{bag},\infty}(\hat{\Sigma}_e) \cdot (T - 1)/(T - k)$. The two routinely disagree by a factor of 2–4. We now show that this gap is an artefact of using the wrong inflation factor, and derive the correct one.

Two Sources of Estimation Noise

The exogenous case has a single source of estimation noise: the k combination weights are estimated from T observations, giving the standard OLS inflation $(T - 1)/(T - k)$.

The endogenous case has an *additional* source: the within-subset weight formula $\hat{w}_{S,i} = \hat{\tau}_i / \hat{A}_S$ depends on estimated precisions $\hat{\tau}_i = 1/\hat{\sigma}_i^2$, so even before any OLS step the weights are suboptimal because $\hat{\Sigma}_e \neq \Sigma_e$. The $k \times k$ submatrix $\hat{\Sigma}_{e,S}$, as a marginal of the $m \times m$ $\text{Wishart}(T - 1, \Sigma_e)$, is itself $\text{Wishart}(T - 1, \Sigma_{e,S})$ — the same distribution as if it had been estimated directly from k series. Applying the Kan–Zhou correction to the k -dimensional portfolio formed from $\hat{\Sigma}_{e,S}^{-1}$ gives:

$$E_{\hat{\Sigma}}[\hat{w}'_S \Sigma_{e,S} \hat{w}_S] = \frac{T - 2}{T - k - 2} \cdot Q^{\text{opt}}(S), \quad (52)$$

with k (the subset size) in the denominator, *not* m (the full pool size). The m in the

original Kan–Zhou formula arises from inverting the full $m \times m$ sample covariance; here only the $k \times k$ submatrix is inverted, so the relevant dimension is k .

Remark 9 (Why $KZ(m)$ is irrelevant for k selection). *Replacing $(T - 2)/(T - k - 2)$ by the constant $(T - 2)/(T - m - 2)$ leaves the argmin of L_k^{bag} unchanged, since multiplying every value of the loss curve by the same constant does not shift its minimiser. Only a k -dependent correction can alter k^* .*

Remark 10 (Relation to OLS inflation and the Granger–Ramanathan regression). *The standard inflation factors for the Granger–Ramanathan (1984) regression setups differ by how many free parameters are counted:*

- **GR3** (unconstrained, no intercept, k slopes): residual $df = T - k$, inflation $(T - 1)/(T - k)$ — the formula used in Section 5.6.
- **GR1** (unconstrained + intercept, $k+1$ parameters): residual $df = T - k - 1$, inflation $(T - 1)/(T - k - 1)$.

Numerically, the KZ correction $(T - 2)/(T - k - 2)$ and the $GR1$ inflation $(T - 1)/(T - k - 1)$ are essentially identical: they differ by at most 0.17% for $k \leq 50$, $T = 200$, and both correspond to counting $k + 1$ effective parameters. The KZ formula derives the same correction from Wishart distribution theory (the exact distribution of the sample inverse covariance), while the $GR1$ inflation derives it from classical degrees-of-freedom counting with an explicit intercept parameter. They coincide because using $\hat{\Sigma}_e$ to compute the k precision-weighted combination weights is equivalent to estimating one additional free parameter beyond the k OLS slopes — the “intercept” of the covariance estimation step. The double inflation formula below can therefore be read as:

$$I^{\text{endo}}(T, k) = \underbrace{\frac{T - 1}{T - k}}_{\text{GR3: } k \text{ combination params}} \times \underbrace{\frac{T - 1}{T - k - 1}}_{\text{GR1} \approx \text{KZ: } k+1 \text{ effective params}} \approx \frac{(T - 1)^2}{(T - k)(T - k - 1)}.$$

The Double Inflation Formula

Combining both sources yields a *conservative double-penalty criterion*:

$$I^{\text{endo}}(T, k) = \underbrace{\frac{T - 2}{T - k - 2}}_{\substack{\text{covariance estimation} \\ (\text{KZ} \approx \text{GR1})}} \times \underbrace{\frac{T - 1}{T - k}}_{\substack{\text{combination weights} \\ (\text{OLS/GR3})}} = \frac{(T - 1)(T - 2)}{(T - k)(T - k - 2)}. \quad (53)$$

For large T this is approximately $[(T - 1)/(T - k)]^2$: the square of the simple OLS inflation. Both the OLS inflation $(T - 1)/(T - k)$ and the KZ factor $(T - 2)/(T - k - 2) \approx (T - 1)/(T - k - 1)$ measure the same underlying phenomenon — finite-sample estimation cost — but at slightly different parameter counts (k vs $k + 1$). Their product is the total inflation for a procedure that estimates *both* the covariance structure and the combination

weights from the same T observations. The combined factor penalises large subsets much more severely than the single-factor formula; Table 6 illustrates the difference at $T = 200$.

Table 6: Inflation factors at $T = 200$: simple OLS vs double correction.

k	$(T - 1)/(T - k)$	$(T - 2)/(T - k - 2)$	$I^{\text{endo}}(T, k)$	Ratio to simple	
5	1.021	1.026	1.047	1.026	
11	1.053	1.059	1.115	1.059	
20	1.106	1.112	1.230	1.112	At $k = 38$,
26	1.144	1.151	1.317	1.151	
38	1.228	1.238	1.520	1.238	
50	1.327	1.338	1.775	1.338	
the double inflation (1.520) is 24% larger than the simple inflation (1.228), making large subsets substantially more expensive.					

Unbiasedness and Correspondence with the Exogenous Criterion

The in-sample risk $Q_k^{\text{bag}}(\hat{\Sigma}_e)$ is a *downward-biased* estimate of $Q_k^{\text{bag}}(\Sigma_e)$:

$$E_{\hat{\Sigma}}[Q_k^{\text{bag}}(\hat{\Sigma}_e)] \approx \frac{T - k - 2}{T - 2} \cdot Q_k^{\text{bag}}(\Sigma_e). \quad (54)$$

Multiplying by the $KZ(k)$ factor exactly de-biases the estimate:

$$E_{\hat{\Sigma}}[Q_k^{\text{bag}}(\hat{\Sigma}_e) \cdot \frac{T-2}{T-k-2}] = Q_k^{\text{bag}}(\Sigma_e). \quad (55)$$

Therefore the corrected endogenous criterion satisfies

$$E_{\hat{\Sigma}}[L_k^{\text{endo}}] = Q_k^{\text{bag}}(\Sigma_e) \cdot \frac{T - 1}{T - k} = L_k^{\text{exo}}, \quad (56)$$

where $L_k^{\text{endo}} = Q_k^{\text{bag}}(\hat{\Sigma}_e) \cdot I^{\text{endo}}(T, k)$. This criterion is not derived from a formal unbiasedness theorem: the Kan–Zhou correction already describes the out-of-sample risk of estimated minimum-variance weights, and multiplying by a second OLS factor is not automatically free of double-counting. We propose it instead as an *approximation criterion* whose motivation is the two-source decomposition above. Monte Carlo validation (Table 7) shows it tracks the RMSFE-optimal k^* substantially better than the single-inflation criterion across all designs, providing empirical justification even without a formal proof.

Monte Carlo Validation

Table 7 reports k^* under all three criteria for the baseline ($m = 50$, $T = 200$) and the m -scaling experiment ($T = 200$, $m \in \{20, 30, 50, 80, 120\}$).

Three findings stand out. First, the double-inflation criterion matches $k^*(\text{RMSFE})$ *exactly* in 6 of 10 cases, compared to 0 of 10 for the single-inflation criterion. Second, the

Table 7: Optimal k^* under three criteria: RMSFE, single inflation L^{curr} , and double inflation L^{endo} . $T = 200$.

DGP	m	$k^*(\text{RMSFE})$	$k^*(L^{\text{curr}})$	$k^*(L^{\text{endo}})$	
<i>Toeplitz, $\rho = 0.8$</i>					
	20	8	12	8	
	30	8	16	8	
	50	12	26	12	
	80	17	44	14	
	120	22	82	18	Bold: $k^*(L^{\text{endo}}) = k^*(\text{RMSFE})$ exactly.
<i>Factor, $r = 2$</i>					
	20	8	14	10	
	30	10	20	12	
	50	14	28	14	
	80	20	47	17	
	120	18	82	18	

$k^*(L^{\text{curr}})$ uses $(T-1)/(T-k)$ only; $k^*(L^{\text{endo}})$ uses $(T-1)(T-2)/[(T-k)(T-k-2)]$. Remaining gap at $m \geq 80$ reflects model misspecification: the Toeplitz/factor structures do not satisfy common loading, so the precision-weighted weights $\hat{w}_{S,i} = \hat{\tau}_i/\hat{A}_S$ are suboptimal for the true Σ_e ; the Wishart result for the $k \times k$ submatrix holds regardless of loading structure for Gaussian errors, but it does not rescue the weight formula.

residual gap under the corrected criterion is at most 4 units across all designs, compared to 4–64 units under the current criterion. Third, the gap grows with m (larger m/T ratio makes the Wishart approximation less accurate) but remains small in the empirically relevant range $m \leq 50$.

Remark 11 (Conservative empirical criterion for endogenous RSM). *The product factor $(T-1)(T-2)/[(T-k)(T-k-2)]$ is the Gaussian/Wishart alternative to the single OLS correction $(T-1)/(T-k)$. It is nearly identical to the Elliott–Liao-style criterion $[(T-1)/(T-k)]^2$ for large T , and to using two separate GR-type inflation factors as described above. We report it as a conservative empirical rule: it tracks the RMSFE-optimal k^* substantially better than the single-inflation criterion in our simulations (Table 7), but it is not a formal unbiasedness correction. Using it requires $T > k + 2$, automatically satisfied for $k \ll T$.*

8.6 T-Scaling and the $\sqrt{T}$ Approximation

The fitted slope of 0.37–0.39 is statistically consistent with the theoretical prediction $1/(\alpha + 1) \approx 0.37$ (at $\hat{\alpha} = 1.7$) and rejects $T^{1/2}$ at the 10% level.

Why $k^* \approx \sqrt{T}$ appears empirically. The endogenous k^* values in Table 8 are numerically close to $\sqrt{T}$ for $T \in [100, 300]$. This is a finite-sample coincidence, not a structural result. The fitted relationship $k^* \approx 2.19 \cdot T^{0.39}$ produces values close to $\sqrt{T}$ over one

Table 8: Endogenous T -scaling: $k^*(\text{RMSFE})$ by T , B4 specification.

Σ_e	$T = 100$	$T = 150$	$T = 200$	$T = 300$	$T = 500$	$T = 800$	Slope	Rejects $T^{1/2}$?
Toeplitz	10	10	14	14	18	22	0.394**	Yes ($t = -2.1$)
Factor	14	14	14	18	26	26	0.367**	Yes ($t = -1.9$)
$\sqrt{T}$	10.0	12.2	14.1	17.3	22.4	28.3	(0.500)	
$2.19 \cdot T^{1/3}$	10.1	11.6	12.8	14.6	17.3	20.3	(0.333)	

Log-log OLS slopes; ** $p < 0.01$. Reference rows show $\sqrt{T}$ and $2.19 \cdot T^{1/3}$ (constant fitted at $T = 200$, Toeplitz).

decade of T because $2.19 \cdot T^{0.39} = \sqrt{T}$ at $T = (2.19)^{1/(0.5-0.39)} = (2.19)^{9.1} \approx 250$, and the two curves do not diverge greatly for $T \in [100, 300]$. At $T = 800$ the divergence is clear: $2.19 \cdot 800^{0.39} = 20.3$ versus $\sqrt{800} = 28.3$; the data give $k^* = 22$, closer to the power-law prediction.

Theoretically, the $\sqrt{T}$ appearance can be understood through the finite-bagging bridge: when a single $\hat{\Sigma}_e$ is shared across all G bags, the common estimation error introduces a residual linear-in- z_k component in the effective excess risk (equivalent to a finite effective $G_{\text{eff}} \sim T/m \ll G$). This pushes the effective scaling exponent above $1/(\alpha + 1)$ toward $1/2$. The observed slope of 0.39 lies between the two limits, consistent with a partial linear component.

Practical recommendation. The RMSFE curve is shallow near the optimum. At $T = 200$, switching from $k = 9$ ($\approx \sqrt{T}$) to $k^* = 14$ reduces RMSFE by approximately 6%. The rule $k = \lfloor \sqrt{T} \rfloor$ therefore captures the bulk of available gains with no estimation of α or C . We use this rule as our empirical default.

8.7 m -Scaling: How k^* Grows with Pool Size

The cube-root formula (43) scales with T but does not explicitly depend on m . In practice, larger pools require larger subsets because the precision-weight distribution is more spread out and estimating combination weights is noisier when m/T is large. Table 9 fixes $T = 200$ and varies $m \in \{20, 30, 50, 80, 120\}$ using the endogenous MC with B4 coefficients ($N_{\text{rep}} = 500$, $K_{\text{RS}} = 500$).

Table 9: m -scaling: $k^*(\text{RMSFE})$ by pool size, $T = 200$ fixed.

Σ_e	$m = 20$	$m = 30$	$m = 50$	$m = 80$	$m = 120$	$\hat{\beta} \ (k^* \propto m^{\hat{\beta}})$
Toeplitz	8	8	12	17	22	0.61 ($R^2 = 0.96$, $p = 0.004$)
Factor	8	10	14	20	18 [†]	0.51 ($R^2 = 0.91$, $p = 0.012$)

Log-log OLS of $\log k^*$ on $\log m$. [†]Factor $m = 120$: slight dip from $m = 80$ likely reflects grid-step noise. Gains stabilise at 10–11% (Toeplitz) and 4–7% (Factor) across m .

k^* rises sub-linearly with m , with log-log slopes of 0.61 (Toeplitz) and 0.51 (Factor) — well below 1 (linear) and above 0.33 (cube-root). Larger pools require larger subsets to cover the precision-weight distribution, but the benefit of scale is tempered by the rising estimation cost as m/T grows.

9 Empirical Application: SPF (Near-Homogeneous Case)

9.1 Motivation and Design

The Survey of Professional Forecasters (SPF) provides a natural test of the theory’s prediction in the homogeneous regime: when $C_\delta \approx 0$, RSM collapses to the equal-weighted mean and should deliver no gains.

Data. Philadelphia Fed SPF, Q1 1985–Q4 2024. Panel: $N_t \approx 30$ –53 respondents per round (median 36 for real GDP, 35 for CPI). Forecasts: each respondent’s own submission; no additional regression step.

RSM. At each round, draw k forecasters without replacement and take the simple average. Since no precision weights are estimated, the infinite-bagged weight converges mechanically to the full equal-weighted mean: $\mathbb{E}_S[k^{-1} \sum_{i \in S} f_i] = (N_t)^{-1} \sum_{i=1}^{N_t} f_i$. The SPF experiment therefore functions as a sanity check.

9.2 Results

Table 10: SPF RSM results ($G = 1000$ bags, $k = \lfloor \sqrt{N_t} \rfloor$).

Series	h	Median N_t	$k^*(\text{oracle})$	RMSFE (Mean)	RMSFE (RSM)	Gain	
RGDP	1	36	3	6.405	6.392	+0.2%	Gain
RGDP	4	36	13	2.208	2.205	+0.1%	
CPI	1	35	10	0.974	0.969	+0.5%	
CPI	4	35	3	1.471	1.470	+0.1%	

$= (L^{\text{Mean}} - L^{\text{RSM}})/L^{\text{Mean}}$. None are DM-significant.

All gains are below 0.5% and none are statistically significant, confirming the theory’s prediction. The SPF is a near-homogeneous panel: professional economists with similar training and information sets produce forecasts with low precision dispersion ($V_\delta \approx 0$), making RSM no better than the simple mean.

10 Empirical Application: GDP Forecasting with FRED-MD

10.1 Data and Setup

Data. 102 real-time quarterly vintages (1999Q3–2025Q1, vintage 18 excluded for data availability). Predictor pool: $m \approx 126$ monthly FRED-MD series, each converted to a predicted quarterly GDP growth rate via a rolling univariate autoregression with lag order $P \in \{0, 1, 2, 3, 4\}$ selected by AIC at each vintage.

RSM combination. At each period, draw k individual forecasts and compute constrained OLS combination weights. $G = 1000$ draws for the main estimate. Default $k = \lfloor \sqrt{T} \rfloor \approx 9$ –14, chosen before observing evaluation-sample outcomes. Robustness: sweep $k \in \{2, \dots, 55\}$ with $G = 200$.

Benchmarks. Equal-weighted mean, Lasso, Ridge, Random Forest.

10.2 Results

Table 11: FRED-MD GDP: bivariate RSM, full sample (102 quarters).

Model	MAFE	Sig.	RMSFE	Sig.
Mean	0.54		1.10	
RSM	0.45	**	0.76	*
Lasso	0.51		0.73	*
Ridge	0.49	*	0.98	
Random Forest	0.55		1.24	

Table 12: FRED-MD GDP: bivariate RSM, excluding COVID-19 (Q2–Q3 2020).

Model	MAFE	Sig.	RMSFE	Sig.
Mean	0.42		0.60	
RSM	0.38	**	0.53	**
Lasso	0.45		0.58	
Ridge	0.39	**	0.54	**
Random Forest	0.41		0.57	*

one-sided Diebold–Mariano test vs Mean.

RSM reduces RMSFE by 31% (full sample) and 12% (no-COVID) relative to the equal-weighted mean; both are statistically significant. The gains are robust: the no-COVID sample, which excludes the two outlier quarters that disproportionately inflate RMSFE, still shows significant improvement.

Oracle k^* and the $\text{RMSFE}(k)$ curve. Sweeping $k \in \{2, \dots, 55\}$ ex post produces a U-shaped $\text{RMSFE}(k)$ curve in both samples, confirming the interior optimum. The

ex-post minimisers are $k^* = 19$ (full) and $k^* = 21$ (no-COVID), approximately $2\sqrt{T}$. These are oracle values selected on the evaluation sample and should not be interpreted as genuine out-of-sample evidence. The fixed rule $k = 9$ ($\approx \sqrt{T}$) costs 0.09 RMSFE units relative to the oracle—a gap unlikely to be significant in a sample of 102 quarters.

Table 13: FRED-MD GDP AR+X RSM results.

Sample	Bivariate		AR+X	
	k^*	min RMSFE	k^*	min RMSFE
Full	19	0.662	19	0.617
No-COVID	21	0.499	16	0.506

The AR+X specification, which augments each autoregression with a FRED-MD predictor, delivers the best full-sample RMSFE (0.617). The no-COVID oracle $k^* = 16$ is closer to the endogenous MC range 11–14 than the bivariate $k^* = 21$, consistent with the AR+X model producing a tighter predictor distribution.

11 Conclusion

We have studied RSM for forecast combination from first principles. The paper’s main contributions are:

1. **Exact theory.** Under the common-loading model we prove: a conservation law governing precision-weighted weight deviations; an exact formula for the infinite-bagged excess risk $R_k^{\text{bag},\infty}$; monotone decrease of $R_k^{\text{bag},\infty}$ in k (under a stated contraction condition, verified numerically); and a first-order condition for the optimal k^* .
2. **Tractable optimum for the average-subset estimator.** A Taylor expansion of $\mathbb{E}_S[1/A_S]$ yields a linear-in- z_k excess risk and an exact square-root optimal subset size $k_{\text{sub}}^* \propto T^{1/2}$. The infinite-bagged excess risk resists a parallel derivation because the requisite conditional harmonic mean has no closed form.
3. **Power-law parametrisation from Monte Carlo.** With known Σ_e , the Monte Carlo shape of $R_k^{\text{bag},\infty}$ is well-described by Cz_k^α ($R^2 > 0.97$). The exponent $\hat{\alpha} \in (1, 2)$ is determined by heterogeneity H : small H gives $\hat{\alpha} \rightarrow 2$ (fast decay, cube-root k^*); large H gives $\hat{\alpha} \rightarrow 1$ (slow decay, square-root k^*). The optimal subset size is $k^* \propto T^{1/(\alpha+1)}$, validated by exact agreement between the formula and the grid-search minimiser.
4. **Tractable endogenous result under Gaussianity.** The Kan–Zhou correction predicts a modest level shift in k^* (5–15%) while preserving the $T^{1/(\alpha+1)}$ scaling

exponent. The large empirical gap between exogenous and endogenous k^* is driven by non-common-loading misspecification, not estimation noise per se.

5. **Empirical evidence.** RSM delivers 10–30% RMSFE reductions in GDP forecasting. The near-homogeneous SPF panel confirms the theory’s zero-gain prediction when $C_\delta \approx 0$.

A Proofs of Supporting Results

A.1 Derivation of $\bar{v}_{k,i}$

From Definition 1, $v_{S,i} = \tau_i/A_S$ for $i \in S$ and $v_{S,i} = 0$ otherwise. The expected weight is

$$\bar{v}_{k,i} = \mathbb{E}_S[v_{S,i}] = \Pr(i \in S) \cdot \mathbb{E}_S \left[\frac{\tau_i}{A_S} \mid i \in S \right] = \frac{k}{m} \cdot \frac{\tau_i g_i(k)}{k} = \frac{\tau_i g_i(k)}{m},$$

confirming (21).

A.2 Variance of A_S under Uniform Subsampling

Under sampling without replacement of size k from $\{\tau_1, \dots, \tau_m\}$:

$$\mathbb{E}_S[A_S] = k\bar{\tau}, \quad \text{Var}_S(A_S) = k \cdot \frac{m-k}{m-1} \cdot \frac{1}{m} \sum_i (\tau_i - \bar{\tau})^2 = k z_k \frac{m}{m-1} \cdot V_\tau,$$

where $V_\tau = m^{-1} \sum_i (\tau_i - \bar{\tau})^2$ and $z_k = 1/k - 1/m$.